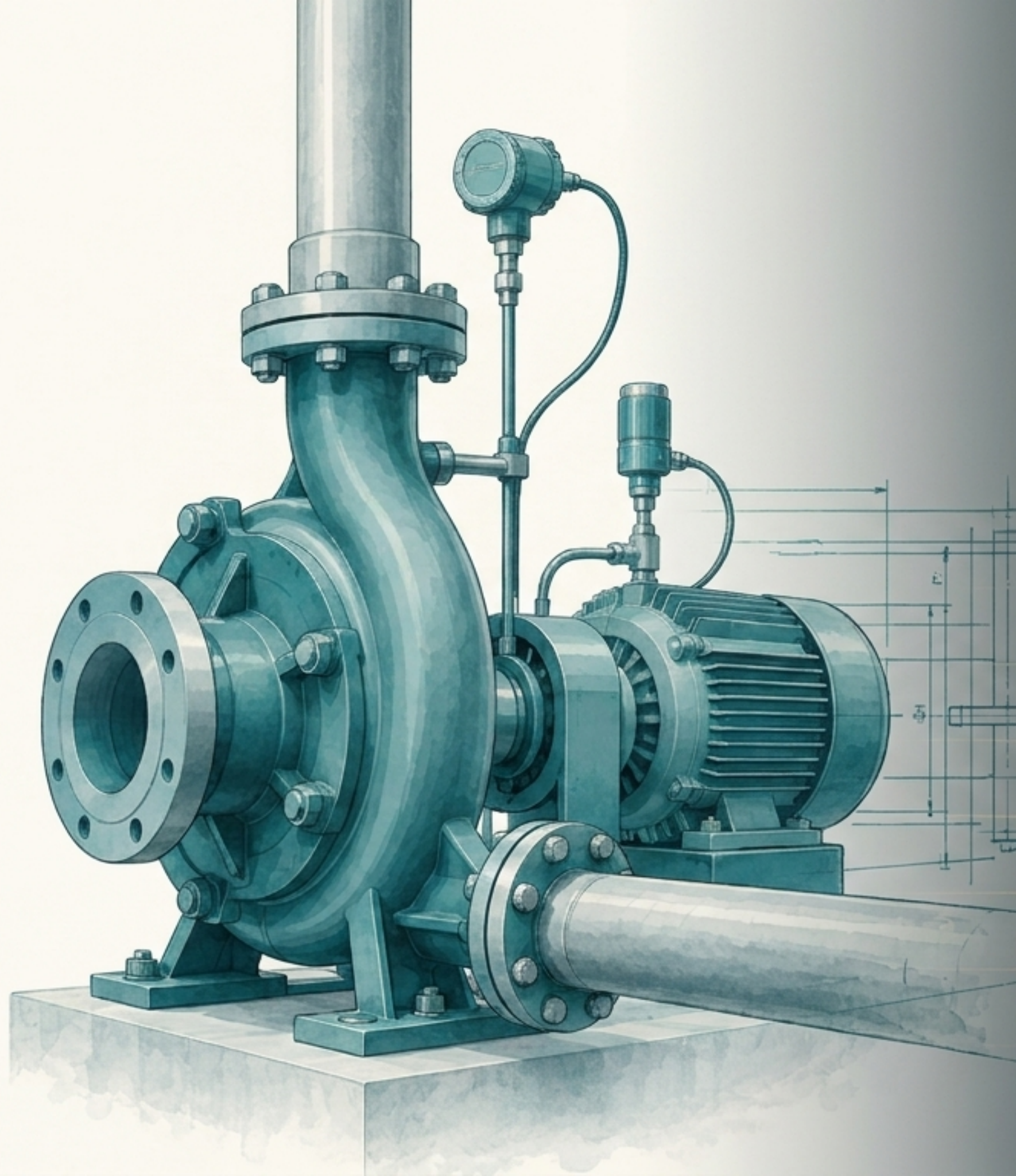
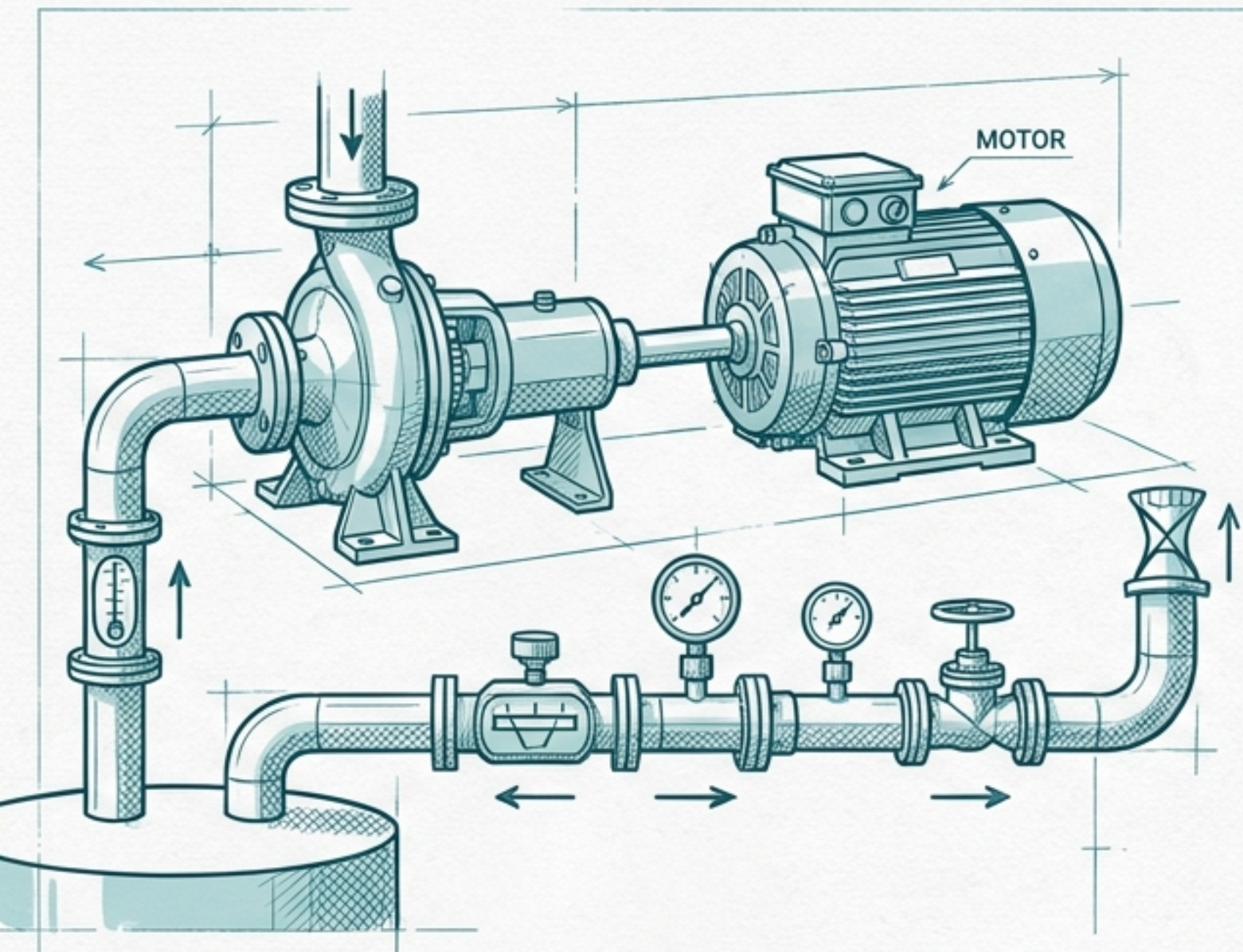


From Physical Pump to Feature Matrix

Translating raw engineering data into machine learning structures.



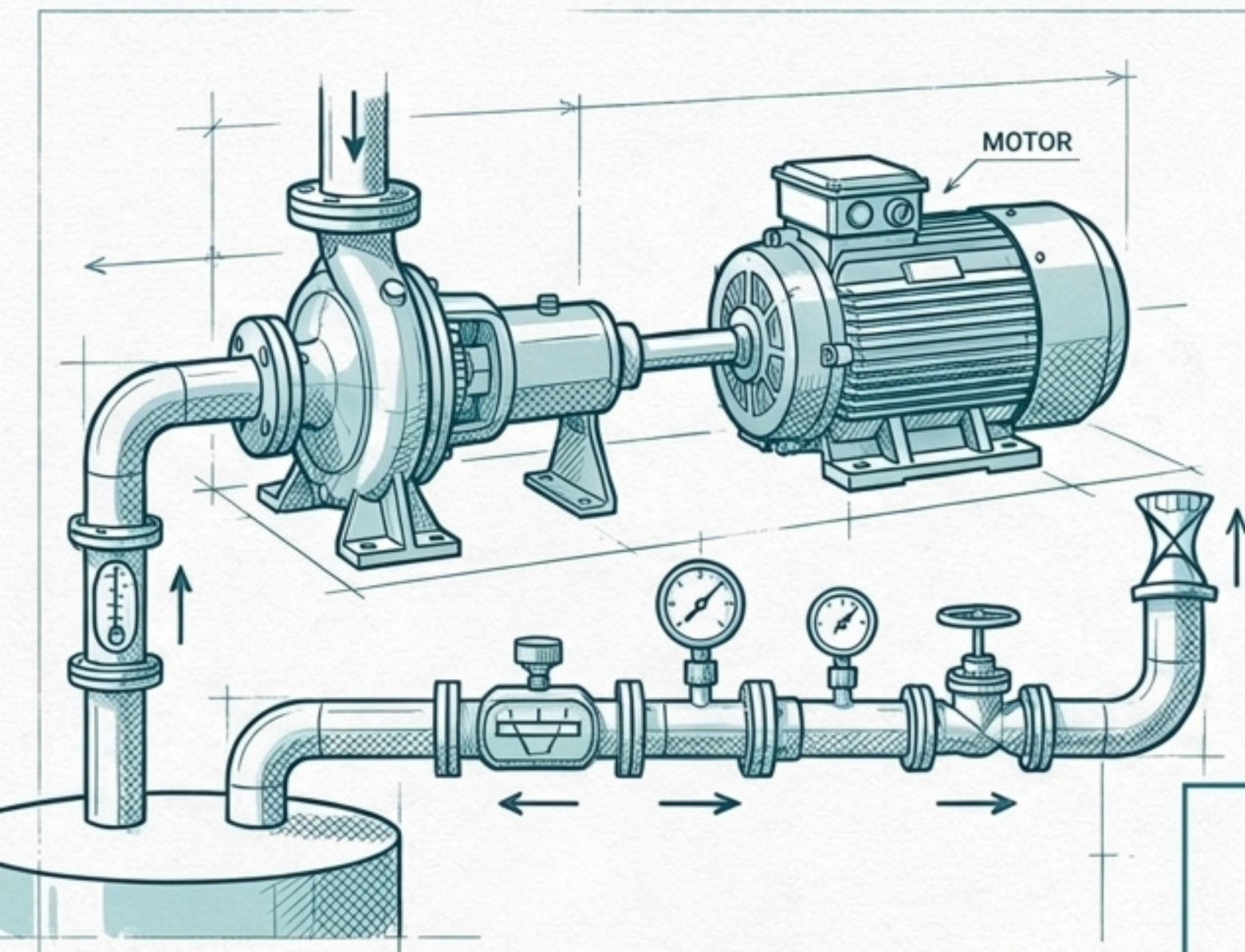
	120.4	45.2	0.98	230.7	1.5	14.2
	-	-	-	-	-	-
3.4	120.4	45.2	0.98	230.7	1.5	3.4
2.6	120.4	45.2	0.98	230.7	3.4	14.2
1.5	95.3	45.2	0.98	230.7	1.2	14.2
6.7	120.4	45.2	0.98	230.7	1.5	3.4
14.2	95.2	45.2	0.98	230.7	3.4	14.2
14.2	95.2	45.2	0.98	230.7	14.2	



The physical reality begins as a raw data table.

Test ID	Flow (m ³ /h)	Pressure Rise (bar)	Speed (RPM)	Electrical Power (kW)	Machine ID
P-001	120.5	4.20	1495	18.5	M-101
P-002	118.0	4.25	1498	18.7	M-101
P-003	121.2	4.18	1500	18.6	M-102
P-004	119.8	4.22	1496	18.4	M-102
P-005	120.1	4.21	1497	18.5	M-103

DATASET DEFINITION: A collection of examples with associated variables and context.
Base scenario: 120 steady-state pump tests.



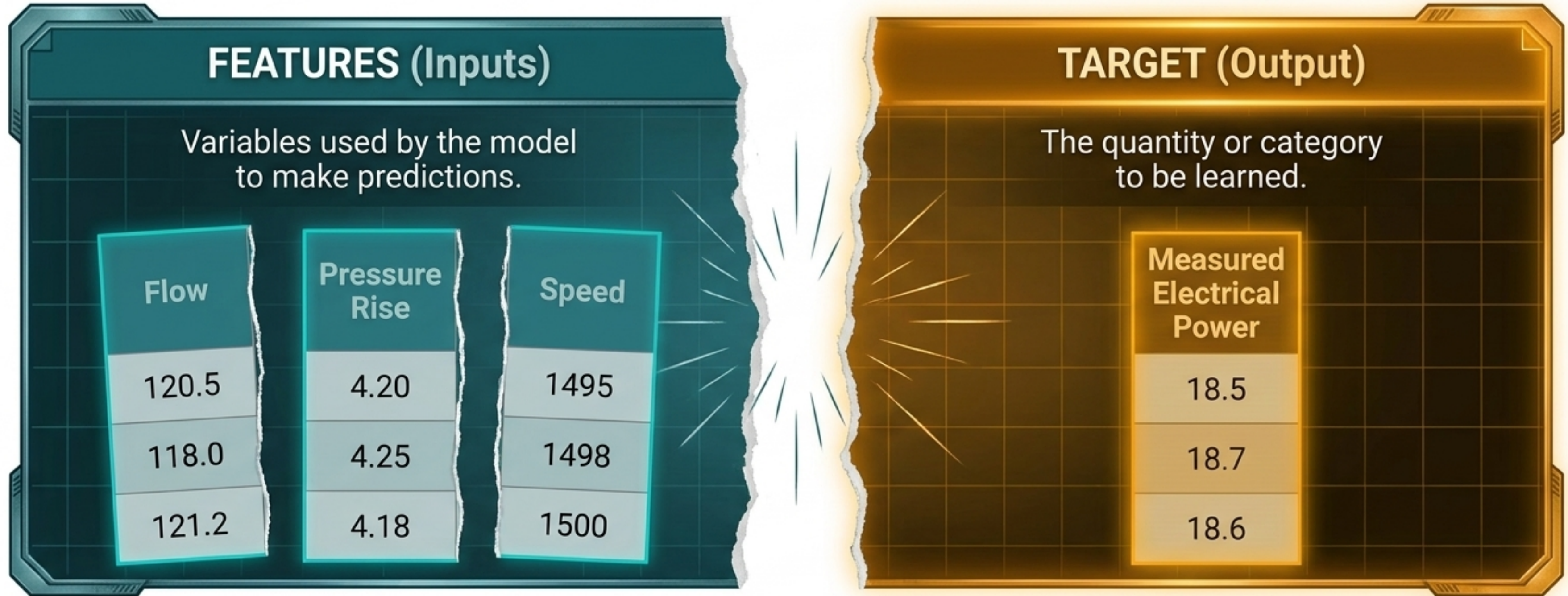
The Dissicting the table into machine learning coordinates.

VARIABLES: The distinct measurements and attributes recorded for every sample.

Test ID	Flow (m³/h)	Pressure Rise (bar)	Speed (RPM)	Electrical Power (kW)	Machine ID
P-001	120.5	4.20	1495	18.5	M-101
P-002	118.0	4.25	1498	18.7	M-101
P-003	121.2	4.18	1500	18.6	M-102
P-004	119.8	4.22	1496	18.4	M-102
P-005	120.1	4.21	1497	18.5	M-103

SAMPLE: One complete example in a learning procicies in learning problem (e.g., one steady operating test).

Separating what we measure from what we want to predict.



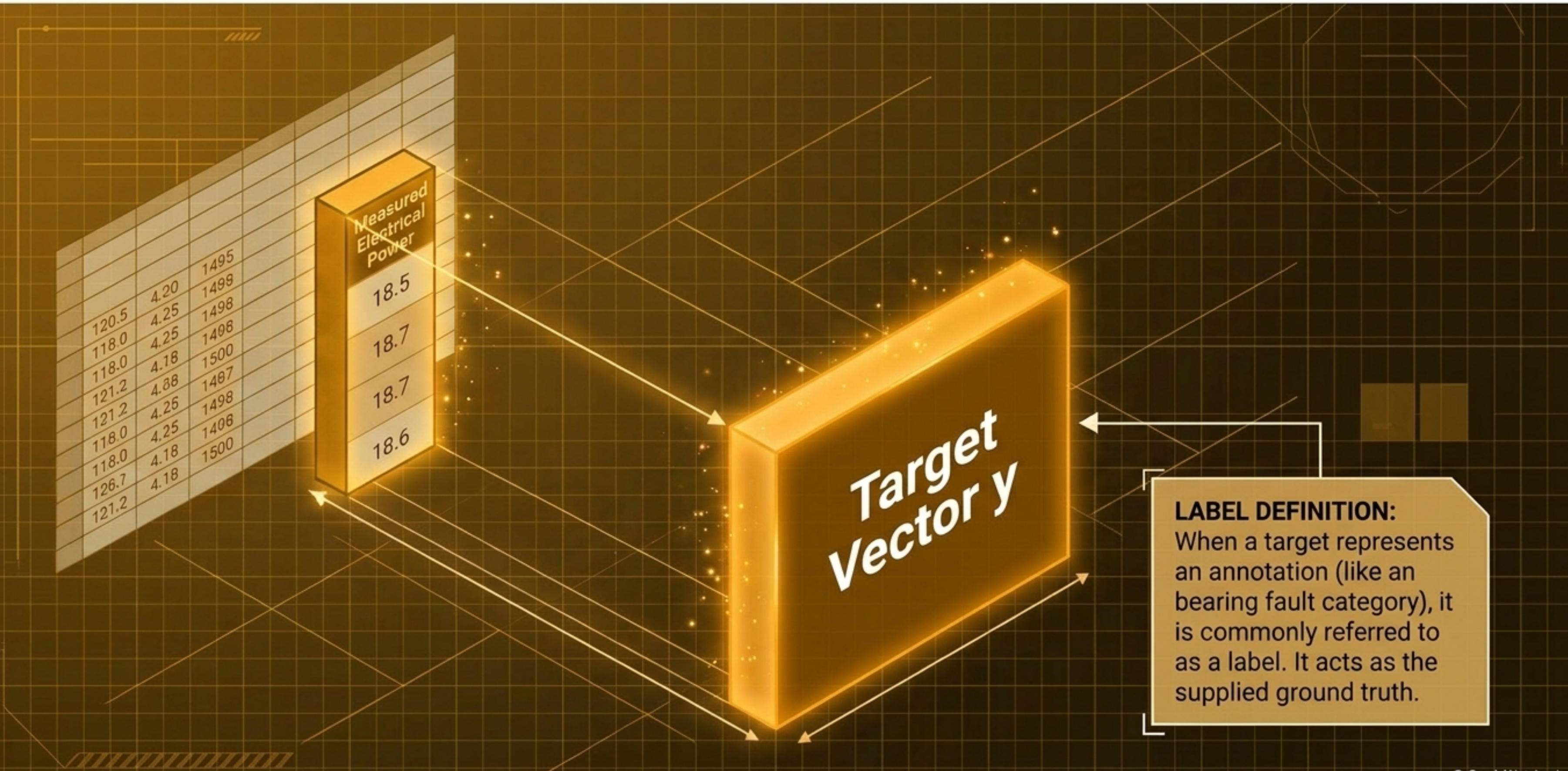
Note: In a different study, electrical power could be a feature rather than a target. Define the decision before assigning columns.

Extracting features into a two-dimensional mathematical matrix.



In regression or classification, all input variables are isolated into Matrix X to isolate predictive signals.

Isolating the ground truth into a one-dimensional target vector.



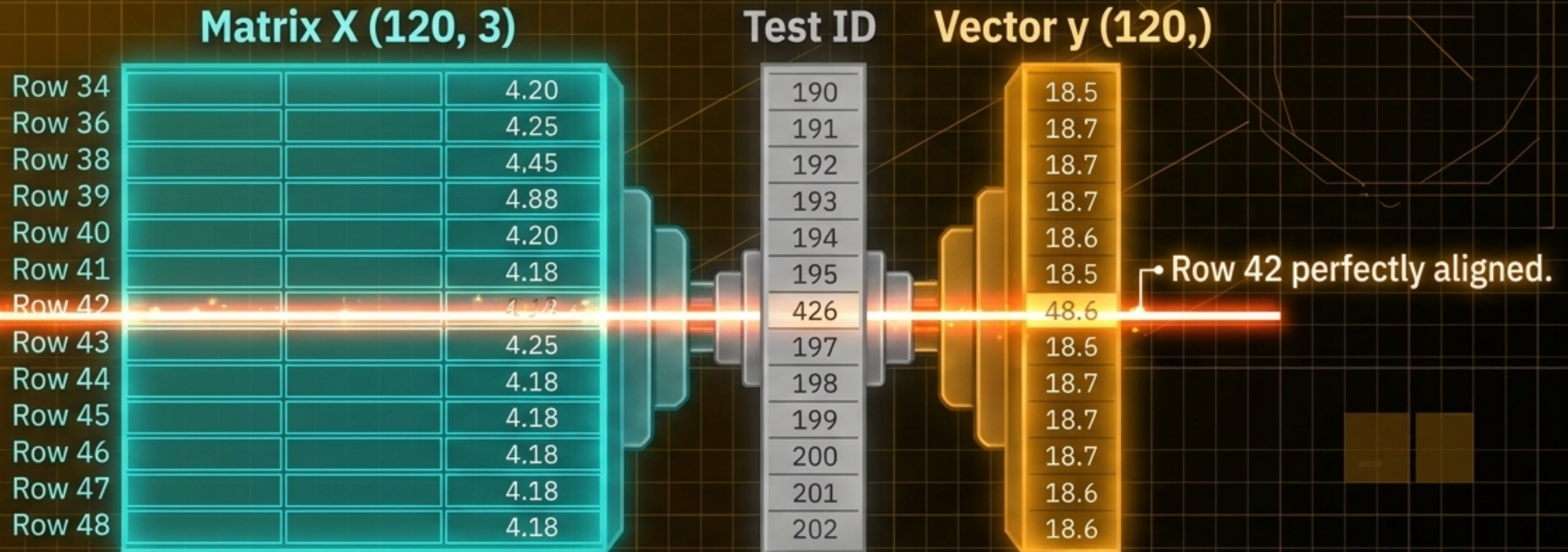
Calculating the exact tensor shapes for 120 independent tests.

X.shape = (n, d)
(120 samples, 3 features)
120 × 3 = 360 total feature values.

y.shape = (n,)
(120 samples,)
120 total target values.

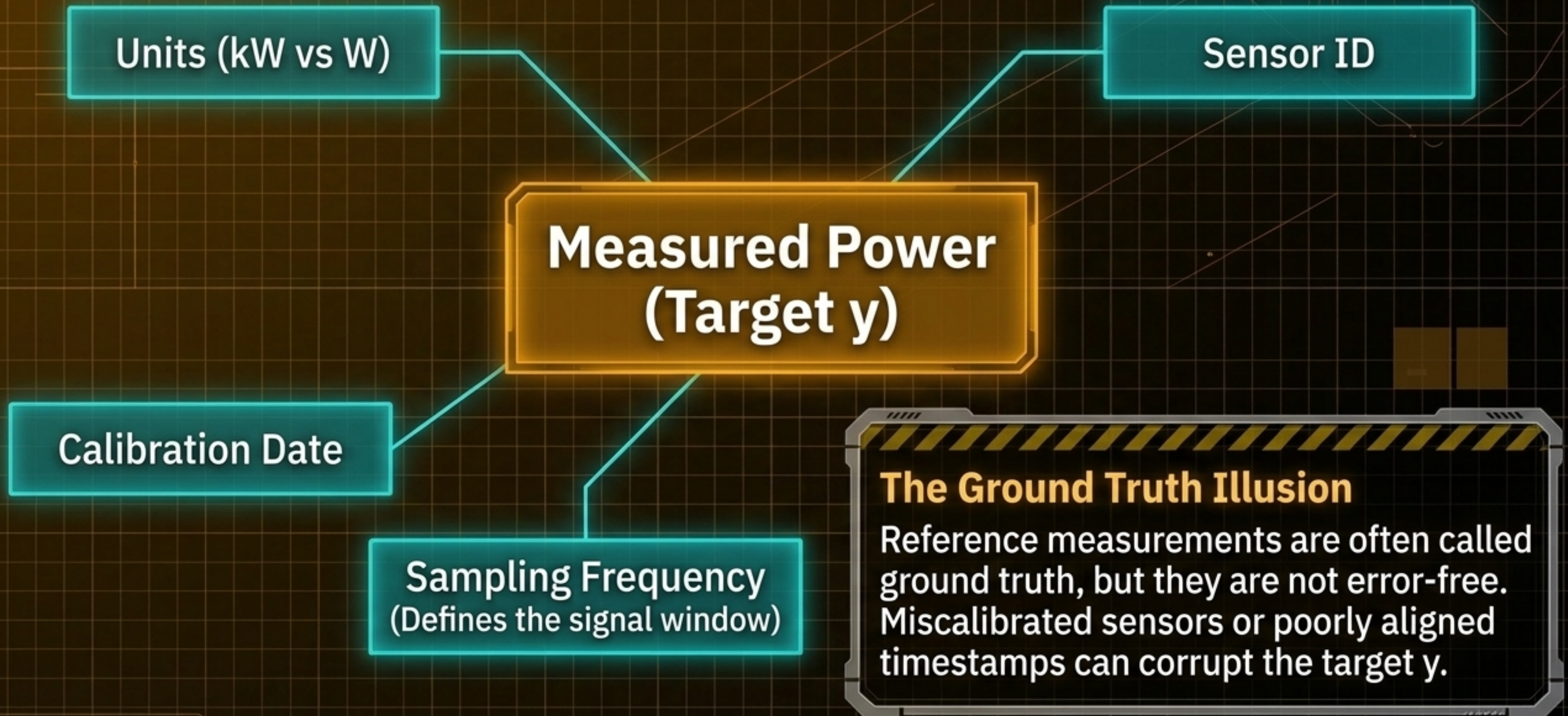
Critical syntax: The trailing comma denotes a 1D scalar array, differentiating rows from columns and preventing accidental misalignment.

Maintaining physical alignment across separated arrays.



The **Test ID** must be stored to keep each scalar target aligned with its specific feature row. Accidental misalignment destroys the causal relationship between input and output.

Numbers require metadata context to carry physical meaning.



Machine ID 12 is not twice as much as Machine ID 6.

Numerical Features

Continuous measurements
or physical counts.

Examples: Flow (m^3/s),
Temperature ($^{\circ}\text{C}$), Pressure (Pa).

Rule: Arithmetic differences and
ratios have explicit physical
meaning.

Categorical Features

Values representing groups,
identities, or states.

Examples: Machine ID,
Material Grade.

Rule: Numeric distances between
IDs lack physical meaning unless
the numbering explicitly encodes
a defined quantity.

The Temporal Rule: Inputs must exist at the time of prediction.

Time of Decision / Alarm Issued

Data available to model

Current Load



Known Speed



Current Vibration RMS



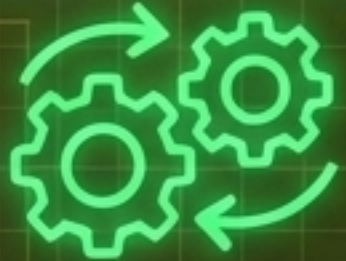
Future data

Post-Failure Repair Code

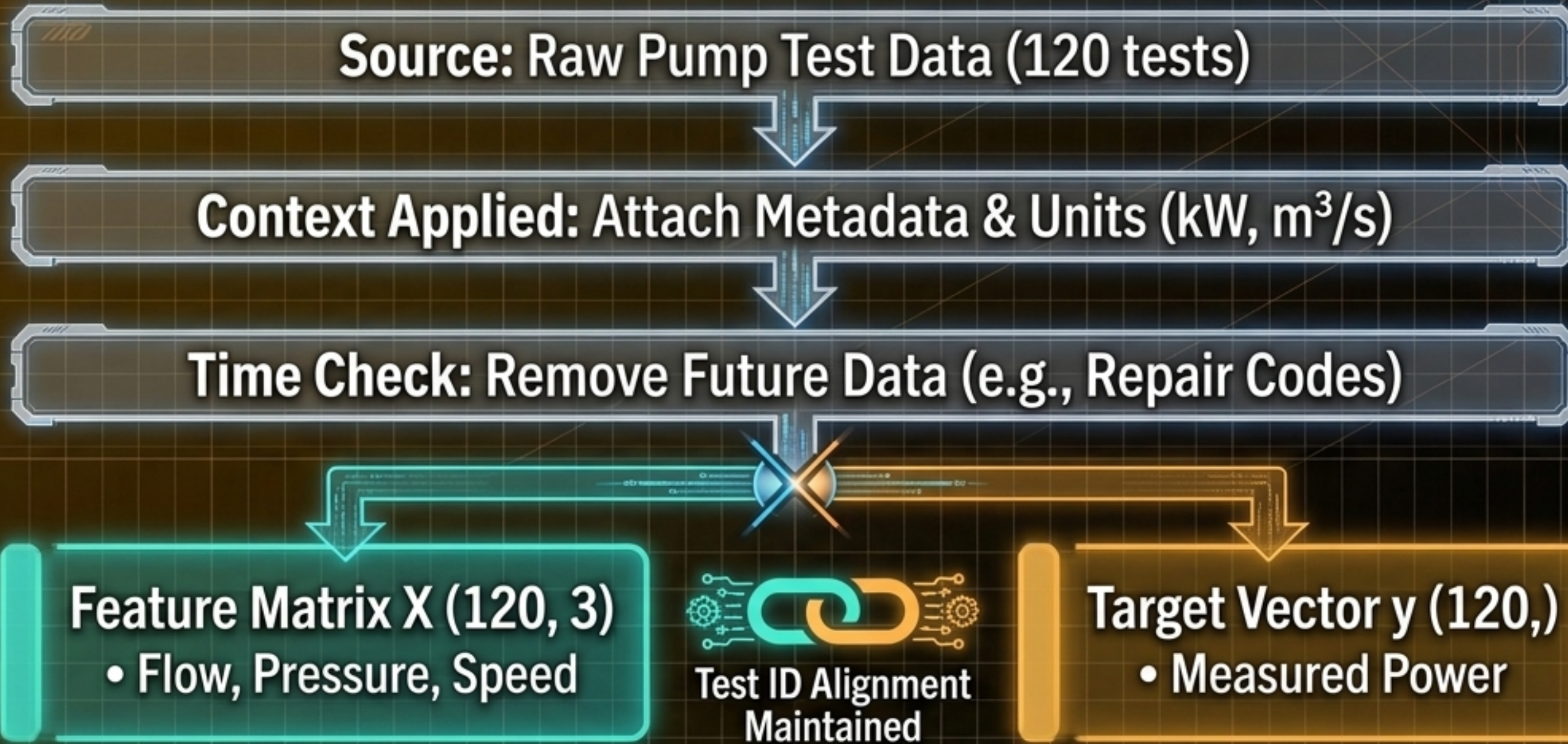


A repair code entered after failure leaks future information into a prediction intended before failure. Using it invalidates the model.

Distinguishing predictive association from physical intervention.

Concept	Question	Engineering Example
 Correlation	Do variables vary together?	Records show Flow and Motor Current increasing simultaneously.
 Prediction	Do inputs estimate unseen outputs?	Can known Flow accurately estimate Current on a new pump test?
 Causation	Does an intervention change the outcome?	Does physically turning a valve to change Flow <i>cause</i> Current to change? (A predictive coefficient alone does not establish this; it requires physics and control conditions).

The Complete Machine Learning Dataset Contract.



A valid predictor is built on mathematically aligned, context-rich data that respects the physical arrow of time.